# LAKSHMAN SIMHA KUPPULA

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## PROFESSIONAL SUMMARY

Entry-level **DevOps / AWS Cloud Engineer** and 2025 B.Tech Computer Science graduate with hands-on experience building cloud infrastructure and DevOps projects. Experienced with **AWS, Linux, Git, GitHub Actions, Docker, Kubernetes, Terraform, Ansible, Bash, Python, Prometheus, and Grafana**. Built end-to-end workflows covering infrastructure provisioning, CI/CD, containerization, deployment, monitoring, and automation. Strong understanding of DevOps principles, Infrastructure as Code, container orchestration, cloud networking, and system troubleshooting.

## TECHNICAL SKILLS

**Cloud:** AWS (EC2, VPC, IAM, S3, RDS, ALB, Auto Scaling, Route 53, CloudWatch, Lambda)
**DevOps:** Git, GitHub, GitHub Actions, Jenkins, CI/CD, Linux
**Containers:** Docker, Docker Hub, Kubernetes, kubectl, Helm, Pods, Deployments, Services, Ingress
**IaC & Automation:** Terraform, Ansible, Bash, AWS CLI, YAML, JSON
**Monitoring:** Prometheus, Grafana, CloudWatch, Logs, Alerts, Incident Troubleshooting
**Programming:** Python

## PROJECTS

**ChaosOps – Cloud DevOps & Self-Healing Platform** *2026*
*AWS | Terraform | Ansible | Docker | Kubernetes | GitHub Actions | Prometheus | Grafana | Python*

- Building an end-to-end DevOps platform demonstrating CI/CD, Infrastructure as Code, containerization, Kubernetes deployment, monitoring, and automation.
- Provisioned AWS infrastructure using **Terraform**, including VPC, public/private subnets, security groups, networking, and EC2 resources.
- Automated server configuration using **Ansible**, including Docker installation and system configuration.
- Designed **GitHub Actions** workflows for automated testing, Docker image builds, registry publishing, and deployment workflows.
- Containerized applications using **Docker** and designed Kubernetes workloads using Pods, Deployments, Services, ConfigMaps, Secrets, and health probes.
- Implementing **Prometheus and Grafana** for infrastructure and application monitoring and incident troubleshooting.

**GitHub: github.com/loki2434/chaosops-platform**

**AWS Multi-Tier Cloud Architecture** *2026*
*AWS | VPC | EC2 | ALB | Auto Scaling | RDS | IAM | CloudWatch*

- Designed a multi-tier AWS architecture using VPC, public/private subnets, EC2, Application Load Balancer, Auto Scaling, and private RDS MySQL.
- Configured **IAM roles, security groups, routing, and subnet isolation** to implement cloud security controls.
- Integrated **CloudWatch monitoring and alarms** for infrastructure visibility and operational alerting.
- Configured Route 53, CloudFront, and AWS WAF as part of the architecture.

**IoT Sensor Simulator – Dockerized REST API CI/CD** *2026*
*Python | Flask | REST API | Docker | GitHub Actions | Docker Hub*

- Built a Python Flask REST API that simulates IoT sensor data.
- Containerized the application using **Docker** for consistent deployment.
- Created a **GitHub Actions CI/CD pipeline** to run tests, build Docker images, and publish images to Docker Hub on code changes.

**GitHub: github.com/loki2434/iot-app-pipeline**

**DevOps Home Lab Automation** *2026*
*Bash | Linux | Docker | Kubernetes | Terraform | Ansible | AWS CLI*

- Developed a Bash automation script to install and configure commonly used DevOps and cloud tools on Linux.
- Implemented `set -euo pipefail`, root-user validation, GPG key verification, and automated service configuration.
- Automated installation of Docker, kubectl, Helm, Terraform, Ansible, AWS CLI, Jenkins, Prometheus, and Grafana.

**GitHub: github.com/loki2434/devops-tools**

## EDUCATION

**B.Tech – Computer Science & Engineering (IoT & Cyber Security)** 2021–2025
PSCMRCET
**Achievements:** 1st Prize – IoT Domain Hackathon; 2nd – Project Expo; 3rd – Prompted Pixel Competition; Silver Medal – Boxing